DOMAIN 4 - EMERGENCY MANAGEMENT (QUESTIONS & ANSWERS)

1425. Under what conditions is Level A protection preferred over Level B protection?

1) Where heat stress is a factor.
2) Where the highest level of skin protection available is needed.
3) Where the highest level of respiratory protection available is needed.
4) Where there is a high potential for fire/explosion.

Ans.2 Level A protection provides a higher level of skin protection than Level B protection.

1426. Water principally extinguishes fire by:

1) Cooling the fuel surface.
2) Excluding oxygen.
3) Pre-wetting the fuel.
4) Dilutuion.

Ans.1 All four answers legitimately describe how water extinguishes a fire. Of these, though, cooling the fuel surface has the most impact.

1427. Building fire safety is achieved by all of the following EXCEPT.

1) Separating igniters from ignitable materials.
2) Managing the impact of fire through building design
3) Taking measures to control products of combustion.
4) Organizing a fire brigade.

Ans.4 Building fire safety may be achieved either by fire ignition prevention or, if a fire does occur, by managing its impact through building design. Ignition prevention is accomplished

by separating igniters from ignitable materials. The products of combustion that must be considered by the building designer can be categorized as thermal (flame and heat) and non-thermal (smoke and gases).

1428. A combustible liquid is best defined by one of the following:

1) Any liquid having a flash point above 100 ^{0}F and below 140 ^{0}F.
2) Any liquid having a flash point at or above 100 ^{0}F.
3) Any liquid having a flash point above 140 ^{0}F.
4) Class II liquids having flash points at or above 100 ^{0}F and below 140 ^{0}F.

Ans.2 A combustible liquid is any liquid having a flash point at or above 100ºF. Flammable and combustible liquids are subdivided into classes. Combustible liquids are: Class II - flash points at or above 100ºF and below 140ºF Class III- flash points at or above 140ºF. NFPA 30 - Flammable and Combustible Liquids Code

1429. A flammable liquid is best defined by one of of the following:

1) Any liquid having a flash point below 73 ^{0}F and boiling point below 100 ^{0}F.
2) Any liquid having a flash point below 100 ^{0}F and vapour pressure exceeding 40 psi(a) at 100 ^{0}F.
3) Any liquid having a flash points at or above 100 ^{0}F and below 140 ^{0}F.
4) Any liquid having a flash point below 73 ^{0}F and boiling point at or above 100 ^{0}F.

Ans.2 A flammable liquid is any liquid having a flash point below 100ºF and having a vapor pressure not exceeding 40 psi(a) at 100ºF. Flammable and combustible liquids are subdivided into classes with Class I having a flash point below 100ºF, Class II having flash points at or above 100ºF and Class III having flash points above 140ºF. NFPA 30 - Flammable and Combustible Liquids Code

1430. Flame spread rating is a measure of a materials' surface burning characteristics. Which test is employed?

1) Time/Temperature curve.
2) Steiner Tunnel Test.
3) ASTM E 32.
4) FM Calorimetry Test.

Ans.2 The 25 ft Steiner Tunnel Test, also known as ASTM E-84, NFPA 255, and UL 723 was developed at Underwriters Laboratories. Due to the need for some method for control of interior finishes, the Steiner Tunnel Test was recognized and adopted by ASTM & NFPA as an official standard in 1958.

1431. The chemistry of fire is depicted by a fire triangle. Which of the following does not apply?

1) Air heat and fuel comprise the three elements of the triangle.
2) All three elements must be present to sustain combustion.
3) The combustion process will cease when any one of three elements is removed.
4) The heating process will cease when the ignition source is removed.

Ans.4 An ordinary fire (extinguished using conventional methods) results from the combination of air (oxygen), fuel, and heat. Take one away and the combustion process will stop.

1432. Tuberculation is:

1) Chemical deposits.
2) Sedimentation.
3) Growth of tuber or other organic matter.
4) Internal pipe corrosion.

Ans.4 Tuberculation is caused by internal pipe corrosion and/or a build-up of rust. Tuberculation cannot be detected without an internal inspection. Build-ups in valve may be an indication that the same is happening within the pipes. Such a build-up can significantly change the volume of water delivered by a system.

1433. One of the key elements of a good fire prevention program in industry is .

1) A no smoking policy.
2) A good housekeeping program.
3) A sprinkler system.
4) Correct storage of flammables.

Ans.2 While all contribute to an effective program, a good housekeeping program can prevent many fires in the workplace. These practices maintain control over many common exposures that can lead to or contribute to the spread of fires.

1434. Overloading an electric circuit is a known fire hazard. This hazard exists due to the heat produced by the overload situation that can be a source of ignition. Overloading a circuit is a greater hazard because.

1) The heat breaks down the electrical insulation which can lead to short circuits.
2) The device connected to the circuit can fail due to a brown-out condition.
3) The circuit breaker can fail.
4) The heat from the circuit can ignite the insulation of the wire.

Ans.1 Overloaded circuits, over time, will cause the insulation to fail, which can lead to short circuits and arcing. Arcing then becomes a significant ignition source.

1435. A command post established for managing a hazardous chemical spill should contain, at a minimum, all of the following EXCEPT:

1) Two radios
2) Spill clean-up equipment
3) Phone
4) Facility MSDS's

Ans.2 A command post is set up away from the spill area, therefore the spill cleanup equipment needs to be at the spill scene, not at the command post.

1436. How often should direct monitoring instruments used for emergency response be checked or calibrated?

1) Quarterly
2) Annually
3) Monthly
4) Before and after each use

Ans.4 Emergency response instrumentation need to be calibrated or response checked (bump tested) prior to and after each use. Instruments should be maintained, calibrated and used in accordance with the manufacturer's instructions. It is critical for responders to use direct monitoring instruments that are operating properly and providing valid results.

1437. What is the most effective component in a fire prevention program for welding or cutting?

1) Inspection of the welding apparatus
2) Upright storage of acetylene cylinders
3) Board-certified welders
4) Cutting and welding hot-work permit system

Ans.4 A permit system which allows safety professionals to preview all cutting and welding work is very successful in eliminating fires.

1438. "Design for building fire safety" represents a systematic approach to the six principal types of fire safety strategies. Three of these are prevention, slowing of initial growth and spread, and detection. Of the following, which is NOT one of the other three strategies?

1) Alarming
2) Suppression
3) Compartmentalization
4) Evacuation

Ans.1 Alarming is not a specific major strategy as defined by the National Fire Protection Association.

1439. Which does a combustible gas indicator best measure?

1) Mercury vapors
2) Lead vapors
3) Petroleum solvent vapors
4) Beryllium oxide

Ans.3 Combustible gas meters, as the name implies, only measure those substances which are flammable. Hence, the correct answer is petroleum solvent vapors.

1440. A flammable gas detected in a confined space registers 10% by volume of the atmosphere. How many parts per million (ppm) does this equal?

1) 100 ppm
2) 1,000 ppm
3) 10,000 ppm
4) 100,000 ppm

Ans.4 To convert any gas concentration to parts per million, simply multiply the percentage expressed as a decimal by 1,000,000. In this problem, 0.10 x 1,000,000 ppm = 100,000 ppm.

1441. A hot wire combustible gas meter should not be used in an atmosphere containing:

1) Flammable liquids
2) Flammable gases
3) Silicone vapors
4) Methane

Ans.3 Silicone vapors rapidly poison the platinum filament in hot wire combustible gas meters. Poisoning can occur when silicone vapors are at very low concentrations (e.g., as low as 1 ppm).

1442. An explosion meter reads 10%. What does this mean?

1) Concentration is 10% of the lower explosive limit

2) Concentration is 10% of the upper explosive limit
3) Concentration is flammable
4) 100,000 ppm flammable gas is present

Ans.1 Explosion meters read in percentage of the lower explosive limit. For example if a compound's LEL were 10%, then a 10% reading on an explosion meter indicates an actual concentration of 1%.

1443. Problem(s) you may encounter with the use of a hot wire combustible gas meter with a platinum catalyst is that the catalyst may become:

1) Inactive
2) Poisoned
3) Enhanced
4) Both 1 and 2

Ans.4 The primary problem with "hot wire" type combustible gas detectors is poisoning of the catalyst. Platinum catalysts are subject to poisoning by silicone vapors at concentrations as low as 1 ppm. This silicone vapor poisoning renders the platinum catalyst inactive.

A secondary problem that is less prevalent today because of advances in meter construction (primarily the coating of the reference bead of the detector with glass), is that un-coated reference beads can become "active" with aging, not inactive.

1444. An instrument that detects a mixture of flammable vapors when air is passed over a heated electric filament is called a:

1) Wheatstone bridge
2) Resistance bridge
3) Combustible gas indicator
4) Gas detector

Ans.3 A combustible gas indicator is the instrument which operates on this principle. A Wheatstone bridge is the actual internal mechanism which provides the instrument with data.

1445. In some combustible gas meters, an electrical circuit consisting of a series of resistors is used to measure the mixture of combustible gas to air. The resistors are balanced and one leg of the circuit, called a hot wire, is exposed to the suspect atmosphere. If a combustible mixture is present, catalytic combustion increases the resistance of the wire and causes an imbalance. The circuit is called a:

1) Combustible resistor circuit
2) Wheatstone bridge
3) Combustible balancing circuit
4) Hot-wire detector

Ans.2 The circuit is called a wheatstone bridge.

1446. A combustible gas meter works on the principle of:

1) A wheatstone bridge
2) Ionization
3) Thermalization
4) Ion disintegration

Ans.1 The combustible gas meter works on the wheatstone bridge principle. A wheatstone bridge is an electronic circuit that is capable of detecting very small changes in current flow via changes in resistance from the heat released from the actual combustion of flammable/combustible materials.

1447. The resistance of a wheatstone bridge _____ when a flammable mixture is oxidized as it passes over a catalyst, releasing heat. The heat causes the resistance in the circuit to increase.

1) Increases
2) Decreases
3) Stays the same
4) Decreases then increases

Ans.1 The resistance increases. Most combustible/flammable gas detectors use this phenomenon to quantify the presence of combustible/flammable materials in an atmosphere.

1448. CHEMTREC is an arm of the Chemical Manufacturers Association (CMA) that provides:

1) Emergency response information
2) Hazard profiles of chemicals in non-emergency situations
3) Cleanup facilities for its members
4) Information system consultants to the semiconductor industry

Ans.1 CHEMTREC is a highly successful program of the CMA to assist in response to HAZMAT incidents.

1449. Fires have various stages of burning. All the following are true EXCEPT:

1) Incipient Stage - No visible smoke, flame, or significant heat is developed, but a significant amount of combustion particles are generated over a period of time
2) Smoldering Stage - The quantity of combustion particles decreases to the point where they become almost invisible
3) Flame Stage - The point of ignition occurs and flames start
4) Heat Stage - Large amount of heat, flame, smoke, and toxic gases are produced

Ans.2 Smoldering Stage - The quantity of combustion particles increases to the point where they become visible (smoke).

1450. In the theory of fire extinguishment, all of the following are believed appropriate EXCEPT:

1) Extinguishment by cooling
2) Extinguishment by oxygen dilution
3) Extinguishment by fuel removal
4) Extinguishment by fire break

Ans.4 Per the NFPA Fire Protection Handbook, all but extinguishment by fire break, are appropriate.

1451. The level of protection that chemical protective clothing offers is affected by all of the following EXCEPT:

1) Durability
2) Permeability
3) Degradation
4) Penetration

Ans.1 Durability is not one of the factors that affect protection levels for chemical protective clothing (CPC).

1452. Of the following, the most effective way to control heat stress is by:

1) Ingesting salt pills
2) Deploying administrative work practices and controls
3) Using personal protective equipment
4) Employing engineering controls

Ans.4 Heat stress is usually controlled by engineering methods, administrative work practices, or by the use of personal protective equipment.

ENGINEERING METHODS are procedures which may be used to reduce the stress of a hot environment. Examples include general ventilation, spot cooling, local exhaust ventilation, air conditioning, cooling fans, and radiant heat shields.

ADMINISTRATIVE WORK PRACTICE controls include acclimatizing to heat, a work-rest program, rotation of workers, education, and training.

PERSONAL PROTECTIVE EQUIPMENT for heat stress is ordinarily used only when a worker must remain in a hot environment long enough to cause unacceptable high heat strain without protection.

1453. A one story steel framed factory building with a metal roof, concrete floor, and masonry exterior/interior walls represents what type of construction?

1) Fire-resistive
2) Non-combustible
3) Ordinary

4) Heavy timber

Ans.2 A standard classification of buildings into five major types based on the fire-resistance ratings of their component parts was developed by NFPA to help prevent confusion of terms used in various building codes and by insurance companies. Non-combustible construction is made entirely of non-combustible materials that either fail to meet the full requirements for fire-resistive construction or are without fireproofing protection for the structural steel elements.

1454. What category, by property class, has the highest incidence of fatal fires (fires that take 50 or more lives per incident)?

1) American community (stores, theatres, clubs, etc.)
2) The industrial environment, excluding mines
3) The industrial environment, including mines
4) The mobile environment (planes, trains, ships)

Ans.3 From 1900, 98 fires caused 50+ deaths. Of these 98, 68 were in the industrial environment, and 59 of these were in mine fires or explosions.

1455. The traditional method of describing fire growth has been through fire loads reflected in:

1) Occupancy classifications
2) Special hazards
3) Flame spread
4) Ordinary hazards

Ans.1 Occupancy classifications include consideration for special hazards and ordinary hazards. Fire load considers flame spread.

1456. Water damage from fire fighting is a major cause of damage from fires. What is the most important consideration to reduce damage from water?

1) The water tightness of the floor
2) The number and location of floor drains
3) The use of less water to fight the fire
4) The use of lifts to raise critical equipment off the floor

Ans.2 Floor drains reduce water damage more than any other approach, especially during salvage operations after the fire.

1457. A tank truck has been involved in a fire. By the time you arrive, the label on the truck only indicates that the material has a pH of 12. What does this tell you?

1) Material is flammable
2) Material is acidic
3) Material is basic
4) Material is neutral

Ans.3 A pH of 12 is a basic solution. A pH less than 7 is acidic, a pH equal to 7 is neutral, and a pH greater than 7 is basic.

1458. In one type of explosimeter, a series of resistors is balanced and a portion of the circuit is exposed to air containing a combustible gas. The combustion causes an imbalance. The circuit is called a:

1) Combustible resistor circuit (CRC)
2) Wheatstone bridge
3) Combustible balancing circuit (CBC)
4) Combustible ampere circuit (CAC)

Ans.2 A common circuit used to measure differences in current flow and hence resistance is the Wheatstone bridge. In this circuit the resistances are balanced, then one part of the circuit

is exposed to an external atmosphere containing flammable gases. This unbalances the circuit because the heat of combustion from the hot wire interacting with the flammable gas results in a measurably higher resistance.

1459. An MFL wall is:

1) An interior partition that acts as a barrier to the spread of fire
2) A fire resistant wall designed to prevent the horizontal spread of fire
3) A self supporting wall designed to maintain its integrity if the structure on either side completely collapses
4) A concrete block, floor to ceiling, wall that protects one fire zone from the other

Ans.3 A maximum foreseeable loss (MFL) wall is a fire resistant wall designed to prevent the horizontal spread of fire into adjacent areas and by insurance carrier decree, must be self supporting and designed to maintain its integrity if the structure on either side completely collapses. The MFL wall insures that a total fire loss would not extend beyond the wall.

1460. If the flowing pressure at a fire extinguishing nozzle is 75 psi, what is the velocity discharge in ft/sec?

1) 90
2) 105
3) 210
4) 4,680

Ans.2 The solution involves two pressure formulas:
$V=\sqrt{2gh}$ - Torricelli's Law for efflux from a container
and
P = 0.433 * h
V = velocity in ft/sec (unknown)
g = acceleration due to gravity (32 ft/sec^2)
P = 75 psi
Solving for "h" (head in feet).
75 psi =(0.433)h
173.2 feet = h

$V=\sqrt{2*32*173.2}$
V = 105.3 ft/sec

1461. Which of the following has the most potential for fire and explosion?

1) Acetone
2) Acetylene
3) Gasoline
4) Hydrogen

Ans.2 The range of flammable limits for acetylene (2.5% to 81% acetylene in air) is greater than that of other commonly used gases, with consequently greater hazard.

1462. Means of egress components in a two-story office building include one 34 in. wide door, a 44 in. wide stairway and a 44 in. wide door. What is the capacity of this exit?

1) 130 persons
2) 147 persons
3) 170 persons
4) 220 persons

Ans.2 The most restrictive egress component is the 44" wide stairway. The component with the least capacity determines the egress capacity.

COMPONENT	CAPACITY FACTOR	CAPACITY
Door 34 in.	0.2 in/person	170 persons
Stair 44 in.	0.3 in/person	147 persons
Door 44 in.	0.2 in/person	220 persons

1463. If the public water supply at street level is 95 psi(g), what is the gauge pressure at a hose outlet located on the roof of a 50 ft high office building?

1) 33 psi
2) 45 psi
3) 73 psi
4) 145 psi

Ans.3 The pressure exerted by a 50 ft high column:
$P = 62.4\ lb/ft^3 * 50\ ft$
$= 3{,}120\ lb/ft^2$
Converting to lb/ft^2 to psi:
$P = 3{,}120\ lb/ft^2 * (1\ ft / 12\ in)^2$
$= 21.7\ psi$
Subtract this from supply pressure at street level
Pressure = 95 psi - 21.7 psi
= 73.3 psi
NOTE: These are gauge pressures.

1464. Class C fires involve:

1) Ordinary combustibles
2) Live electrical equipment
3) Combustible metals
4) Flammable liquids

Ans.2 Class A fires involve ordinary combustibles such as paper and cardboard. Class B fires involve flammable or combustible liquids. Class C fires involve live electrical equipment. Class D fires involve combustible metals.

1465. A motor rated to operate in a flammable/explosive atmosphere is explosion proof due to:

1) It's ability to contain internal explosions
2) It's being constructed exclusively of bronze
3) It's ability to eliminate arcs
4) It's ability to operate at the non-sparking point of the fan curve

Ans.1 Explosion proof motors are designed to prevent explosions occurring inside the motor from propagating to the external environment.

1466. At a construction site, an approved/rated storage cabinet may hold no more than _____ gal of flammable materials or _____ gal of combustible materials.

1) 30, 60
2) 60, 120
3) 120, 240
4) 5, 10

Ans.2 An approved/rated cabinet may hold up to 60 gal of flammable materials or 120 gal of combustible materials. No more than three such cabinets may be located in a single storage area.
29 CFR 1926.152 (b)(3)

1467. Chlorinated hydrocarbon solvents involved in a fire may generate:

1) Arsine
2) Phosphine
3) Phosgene
4) Styrene

Ans.3 Phosgene ($COCl_2$) may be generated during the thermal decomposition of chlorinated hydrocarbons.

1468. Indicate the flammable solvent:

1) Mono-chlorobenzene
2) Freon TF
3) Trichloroethylene
4) 1,1,1-trichloroethane

Ans.1 Chlorobenzene is a flammable solvent. The other solvents are chain hydrocarbons with chlorine substitution. Normally, this indicates a tendency towards non-flammability, but in this case a chlorinated solvent is also a flammable solvent.

1469. One class of organic chemicals that form potentially explosive peroxides is:

1) Ethers
2) Ketones
3) Alkyls
4) Anilines

Ans.1 Aged ethers can form explosive peroxides. These peroxides may frequently occur as visible crystal growths in the ether solution.

1470. Which of these metals is combustible?

1) Nickel
2) Magnesium
3) Molybdenum
4) Iron

Ans.2 Examples of metals which are considered combustible other than magnesium include titanium, zirconium, sodium, and potassium. Fires involving these metals are Class D fires.

1471. A fire protection standpipe 110 ft high is filled with water. What is the gauge pressure at the bottom of the pipe?

1) 33 psig
2) 38 psig
3) 48 psig
4) 62 psig

Ans.3 To determine the pressure at the bottom of the pipe, obtain the density of water from the OHST/CHST Examination Reference Handout and multiply it by the height of the standpipe. Convert the units to obtain pounds per square inch (psi).

Water has a density of 62.4 lbs./ft^3.

110 ft (62.4 lbs./ft^3)(1 ft/12 in.)^2 = 47.67 psi

To obtain the absolute pressure, add 14.7 psi to the gauge pressure.

1472. Fire loads depend on the occupancy and the materials used in their operations. An occupancy that contains mostly paper and wood, such as an office, would have a heat of combustion of:

1) 8-10,000 Btu/lb
2) 7-8,000 Btu/lb
3) 5-8,000 Btu/lb
4) 1-5,000 Btu/lb

Ans.2 A fire load of 7,000 to 8,000 Btu/lb can be expected in this environment.

1473. If a fire pump is to obtain water by the suction process, how far can the pump be expected to raise the water from its source to the pump?

1) 10 ft
2) 25 ft
3) 40 ft
4) 50 ft

Ans.2 Regardless of pump size, 25 ft is about the limit a pump can raise water by the suction process.

1474. When reviewing the effectiveness of compartmentalization as a delay or prevention of the propagation of fire from one area to another, you must consider inherent fire resistance, the details of construction, and:

1) The alarm system in use
2) The speed of reaction of the suppression system
3) The fire codes
4) The penetrations of the walls and floors

Ans.4 One of the major causes of propagation from one area to another are the penetration of walls and floor such as doors, ducts, stairs, utility chases, etc.

1475. The fire point is the lowest temperature at which a liquid in an open container will evolve vapors fast enough to support continuous combustion. That temperature is usually:

1) The flash point
2) A few degrees below the flash point
3) A few degrees above the flash point
4) The ignition temperature

Ans.3 The fire point is usually slightly above the flash point in order to create a mixture between the lean limit and the rich limit thereby creating a flammable mixture.

1476. A catastrophic hazard in large steel foundries that utilize below the floor casting pits is:

1) Loss of power in the electrical system during a pour
2) Explosions from water in the pit that turns to steam when molten metal is poured into the mold
3) Pouring metal into a collapsed mold
4) Cracked ladles

Ans.2 While all are significant hazards, water in a mold cavity can cause a significant explosion when the molten metal is poured into the mold cavity.

1477. The number of hours during which the required fire flow should be available varies from:

1) 1 - 4 hr
2) 2 - 4 hr
3) 2 - 10 hr
4) 3 - 8 hr

Ans.3 The time varies from 2 to 10 hr, although most water authorities place an upper limit of 2 to 4 hr on water supply duration because of economics.

1478. You have a need to determine the rate of flow of water from hydrants in the area of your building. You have been told by the water department that the pressure, or velocity head, of the hydrant in front of your building is 49 psi when measured at the 2.5 inch hydrant butt with a pitot tube and gauge. The k factor for this type of hydrant is 167.8. What is the water volume?

1) 834 gpm
2) 1,175 gpm
3) 1,468 gpm
4) 1,600 gpm

Ans.2 $Q = k\sqrt{P}$
$= 167.8\sqrt{49}$
$= 167.8 * 7$
$= 1,174.6$ gpm

1479. An alarm system that transmits water flow alarms from sprinklers and supervisory signals to a central system that is constantly staffed by the owner's security force is the description of a:

1) Proprietary system
2) Central station system
3) Auxiliary system
4) Remote station system

Ans.1 A proprietary system is a normally closed-circuit system with electromechanical devices used to transmit signals to the central supervisory station located in or near the building or groups of buildings being protected. The central supervisory station is constantly staffed by qualified personnel hired by the building owner or occupant.

1480. Smoke detectors work best where:

1) Smoke generation will be difficult to detect
2) Smoke will be generated by the products of combustion
3) There is the potential for substantial fumes to be generated
4) The products of combustion generated by fire are invisible

Ans.2 Smoke detectors respond to the particles of combustion that occur in the second fire phase. Smoke detectors operate on lighting principles. When smoke enters a light beam, it triggers a reaction; the more smoke that enters, the faster the response.

1481. The main disadvantage of most heat detectors is:

1) Their false alarm rate is high
2) They are costly to install
3) They are slow in detecting fire
4) They have limited application

Ans.3 Of the four kinds of fire detectors in use, heat detectors are the slowest in detecting fires. They are widely used because they are the least expensive fire detector and have the lowest false alarm rate.

1482. Which of the following smoke detectors will accurately detect most fires more rapidly than the others?

1) Rate-of-rise thermal
2) Ionizing
3) Reflected beam photoelectric
4) Bulb detection system

Ans.2 Ionizing detectors sense fire at the earliest practical detection stage. They sense both visible and invisible products of combustion which enters a chamber containing positive and negative plates and a very minute amount of radioactive material that ionizes the air in the chamber. The movement of ions between plates sets up a small current that triggers an alarm.

1483. Smoke detectors which detect the products of combustion are known as what type of detector?

1) Non-ionizing
2) Ionizing
3) Photo-electric
4) Combustion

Ans.2 Combustion produces products that can be detected by the ionizing detector before smoke is visible.

1484. Smoke detectors which react to light reflected off smoke from a fire are known as what type of detector?

1) Non-ionizing
2) Ionizing
3) Photo-electric
4) Combustion

Ans.3 Photo-electric detectors essentially work when smoke passes through the detector and light is reflected back into the photo cell of the detector.

1485. A water motor gong is powered by:

1) Domestic water lines opened electrically by a smoke detector
2) Electrical power
3) A fire pump
4) Water flowing through the sprinkler system

Ans.4 Water flowing through a sprinkler system will power the gong, whether the sprinkler is actuated by a fire or has a substantial leak.

1486. The oldest type of automatic fire detection device in use today is the:

1) Eutectic metal heat detector
2) Ionizing smoke detector
3) Dissimilar metal heat detector
4) Photo-electric smoke detector

Ans.1 Eutectic metals are designed to melt (fuse) at a specific temperature. A fusible link is one of the oldest types of heat detectors, and the heat detector is the oldest type of automatic

fire detection device. It is the type of fire detection used in most sprinkler heads, auto-closing fire doors, automatic lid closers, and some alarms. The development of sprinklers began in the 1860's.

1487. Eutectic metals are used in:

1) Smoke detectors
2) Metal flues
3) Heat detectors
4) Hazardous environments requiring non-sparking tools

Ans.3 Eutectic metals are metal alloys designed to melt at a specific temperature. They are usually made from tin, lead, cadmium, or bismuth, and melt rapidly at a predetermined temperature.

1488. The type of smoke detector that contains a small amount of radioactive material as part of its operating mechanism is the:

1) Light obscuring detector
2) Cloud chamber detector
3) Ionizing detector
4) Light scattering type

Ans.3 Ionizing detectors use radiation to ionize smoke particles so that the particles conduct electrical current, thus operating the alarm. The other types listed are photo-electric smoke detectors that use light as the basis of their operation.

1489. In a pre-action sprinkler system, the water is:

1) Admitted to the piping by the use of a pressure actuated valve
2) Contained in the piping system similar to that of a wet system
3) Admitted to the piping through an automatic water control valve that has been activated by a zoned heat or smoke detector
4) Held back from the sprinkler piping by a deluge valve

Ans. Pre-action systems are designed to protect properties where there is danger of serious water damage as a result of damaged automatic sprinkler heads or piping. The pre-action system is similar to a standard dry-pipe system. To it is added a water supply valve that is opened by the operation of an automatic fire detection system (smoke or heat detectors).

1490. What fire extinguishing agent contains halogenated hydrocarbons?

1) Halon
2) Carbon dioxide
3) Dry powder
4) Foam

Ans.1 Halon is composed of halogenated hydrocarbons (CFC's) which react with stratospheric ozone.

1491. What type of closed-head sprinkler system requires a fire detection device in the affected area to activate?

1) Dry-pipe system
2) Pre-action system
3) Deluge system
4) Wet pipe system

Ans.2 The best answer is a pre-action system. A dry-pipe system relies on sprinkler head operation to trigger water flow. A deluge system uses a fire detection device but the heads are already open.

1492. The flowing pressure at a fire extinguishing nozzle is 50 psi. What is the velocity of discharge in ft/sec?

1) 8.5 ft/sec
2) 50.5 ft/sec
3) 85.5 ft/sec
4) 650 ft/sec

Ans.3 Use the formula $Q = 12.1\sqrt{P}$

$Q = 12.1\sqrt{50}$
$= 12.1 * 7.07$
$= 85.5$ ft/sec

1493. All of the following are decomposition products of Halon 1301 EXCEPT:

1) Hydrogen fluoride
2) Hydrogen bromide
3) Bromine
4) Ethylene

Ans.4 Ethylene is a gaseous olefin compound resulting from the cracking of either selected petroleum fractions or natural gas.

1494. All the following secondary hazards should be considered as a result of discharge of Halon 1301 EXCEPT:

1) Dislodgment of objecte
2) Dislodgment of objects
3) Frostbite burns to skin
4) Heat

Ans.4 Heat is not a secondary hazard as a result of discharge of Halon 1301.

1495. What is one of the major decomposition products produced when Halon is used as an extinguishing agent.

1) Carbon monoxide
2) Hydrogen fluoride
3) Hydrogen chloride
4) Chlorine gas

Ans.2 Hydrogen fluoride is produced when Halon ($BrCF_3$) is exposed to temperatures over 900ºF.

1496. Halogenated hydrocarbons appear to be effective in suppressing fires by:

1) Excluding heat from the fire
2) Producing free radicals that consume free radicals that the fire is producing
3) Cooling the fire
4) Suppressing the production of free radicals by the heat of the fire

Ans.2 These seem to work by producing free radicals that then combine with free radicals being produced by the fire. The elimination of these free radicals then starve the fire.

1497. Water is a particularly effective extinguishing agent because:

1) Water is the cheapest of the extinguishing agents available
2) Water can be used universally
3) Water has a high latent heat and expands 1,700 times when vaporized thereby excluding oxygen
4) Water acts faster than other extinguishing agents

Ans.3 Both of these characteristics make water effective in fighting most fires.

1498. Which of the following extinguishing systems or extinguishing agents are frequently used in areas with electrical hazards (such as switch rooms in manufacturing plants)?

1) Foam Systems
2) Halon systems
3) Dry pipe systems
4) Carbon dioxide systems

Ans.4 Carbon dioxide (CO_2) systems are primarily used because they will not create additional fire hazards or damage the electrical equipment.

1499. A fire emanating from a compressed gas cylinder can be effectively halted _____ .

1) With a handheld extinguisher
2) With a water cannon
3) By shutting off the flow
4) With freon injection

Ans.3 The only way to really halt a compressed gas fire is to shut off gas flow. Temporarily extinguishing the fire may be dangerous if flow is not stopped. The unburned gas can then accumulate and possibly explode.

1500. The handling of ethyl ether involves severe explosion hazards because of its:

1) High flash point
2) Low vapor pressure
3) Ability to form peroxides
4) Both 1 and 2

Ans.3 Ethyl ether has a LOW flash point and a HIGH vapor pressure. Ethyl ether has a dangerous propensity to form highly unstable explosive peroxides when in contact with air

1501. On pressure demand SCBA's there is an alarm to let the wearer know that the air supply has decreased to _____ of full.

1) 10%
2) 20% - 25%
3) 30%
4) 50%

Ans.2 All pressure demand units are required by NIOSH-MSHA to have an alarm to alert the user that the air supply has decreased to 20-25% of full.

1502. The "hot zone" is:

1) The area closest to a radiation leak
2) The area around a spill/release into which untrained personnel may not enter
3) The area where responders and most equipment are decontaminated before entering the "cold zone”
4) The area where only firefighters can enter

Ans.2 The "hot zone" (exclusion zone), is that area around a spill/release into which untrained, unequipped individuals may not enter.

1503. In responding to a spill of a material that is labeled "ORM-A ", what would be your main concern?

1) Asphyxiation
2) Fire/explosion
3) Toxicity
4) Dust

Ans.3 ORM-A refers to a Department of Transportation (DOT) classification of "other regulated materials" that do not meet the definition of other DOT hazard classes. ORM-A materials have anesthetic, irritating, noxious, toxic, or other similar properties that can cause extreme annoyance or discomfort to passengers or crew in the event of leakage during transportation.

1504. A hazardous location, because of the presence of combustible dust requires electrical protection that meets the requirements of:

1) Class I locations
2) Class II locations
3) Class III locations
4) Class IV locations

Ans.2 Class II locations are those in which suspended combustible dust may be present in quantities sufficient to produce explosive or ignitable mixtures; or be electrically conductive; or be ignitable by a malfunction of electrical equipment. NFPA 70 Article 500-6 (National Electric Code)

1505. When applied to electrical apparatus, the term "explosionproof" means:

1) Capable of withstanding external explosions
2) Capable of excluding the intrusion of explosive gas or vapor
3) Capable of containing an internal explosion
4) Both 1 and 2

Ans.3 NFPA 70 (National Electric Code) defines "explosionproof apparatus" as, "Apparatus enclosed in a case that is capable of withstanding an explosion of a specified gas or vapor that may occur within it and of preventing the ignition of a specified gas or vapor surrounding the enclosure by sparks, flashes, or explosion of the gas or vapor within, and that operates at such an external temperature that a surrounding flammable atmosphere will not be ignited thereby." NFPA 70 Article 100-A (National Electric Code).

1506. You are a safety professional in a plant that manufactures a flammable, solvent based ink. The wiring in the processing area should be:

1) Class I, Division 1
2) Class I, Division 2
3) Class II, Division 1
4) Class II, Division 2

Ans.1 Wiring complying with Class I, Division 1 is required. This high level of protection is required because flammable solvents are used. The vapors can travel and find a source of ignition when inadequate protection is installed. NFPA 70 Article 500-5 (National Electric Code)

1507. If a spill occurs in an enclosed space, and the maximum airborne concentration of the spilled material reaches 18,000 ppm, how many room air changes would be needed to drop the airborne concentration of the spilled material to its TLV-C of 1.0 ppm?

1) 4.3 air changes
2) 9.8 air changes

3) 15 air changes
4) 19 air changes

Ans.2 $C = C_i^{(-Vw/Vi)}$
$Vw/Vi = \ln [C_i / C]$
$C_i = 18{,}000$ ppm
$C = 1.0$ ppm
$Vw/Vi = \ln [18{,}000 / 1]$
$= 9.8$

1508. Which of the following temperature ranges concerning fire sprinkler heads color coding is correct?

1) 100ºF Ceiling Temperature - Orange
2) 150ºF Ceiling Temperature - White
3) 375ºF Ceiling Temperature - Blue
4) 475ºF Ceiling Temperature – Uncolored

Ans.2 Use the following table:

Ceiling Temperature (ºF)	Rating	Class	Color
100	135-170	Ordinary	Uncolored
150	175-225	Intermediate	White
225	250-300	High	Blue
300	325-375	Extra High	Red
375	400-475	Very Ex. High	Green
475	500-575	Ultra High	Orange

NFPA 13 2-2.3.2 - Standard for the Installation of Sprinkler Systems

1509. According to NFPA, water fire extinguishers require periodic hydrostatic testing:

1) Annually
2) Every 3 years
3) Every 5 years
4) Every 7 years

Ans.3 NFPA requires hydrostatic testing of water extinguishers every 5 years. The NFPA Fire Protection Handbook is a comprehensive reference covering all aspects of fire extinguishers. NFPA 10 - Standard for Portable Fire Extinguishers

1510. The major cause of fire-related deaths is attributed to:

1) Inhalation of smoke
2) Seared lungs
3) Carbon monoxide
4) Burn trauma

Ans.3 Statistics maintained by the National Safety Council (NSC) and the National Fire Protection Association (NFPA) show carbon monoxide to be the leading cause of fire-related deaths. Carbon monoxide, a gas, inhibits blood cells from carrying oxygen to the body tissues.

1511. What is the leading source of industrial fires?

1) Smoking
2) Lightning
3) Electrical malfunctions
4) Hot surfaces

Ans.3 In an analysis of 25,000 industrial fires reported to FM from 1968-77, electrical malfunctions were the leading cause of industrial fire.

1512. Which activity requires the higher level of personal protection?

1) Spill containment
2) Spill stoppage
3) Decontamination
4) Spill clean-up

Ans.2 The potential exposure during spill stoppage is greater since the situation is not controlled and the responder has to go to the source of the spill, which may have the greatest exposure potential.

1513. EPA specifies levels of protection for chemical protective clothing. One ensemble includes liquid-splash protective suit, pressure-demand full face piece SCBA, inner chemical resistant gloves, chemical resistant safety boots, two-way radio, hard hat. This is an example of what level?

1) A
2) B
3) C
4) D

Ans.2 Level A is more stringent than Level B in that a vapor protective suit is required instead of a liquid-splash protective suit.

1514. The products produced when water is electrolyzed are:

1) H_2O_2
2) H_2 and O_2
3) H+ and OH-
4) H_2O_2 and H+

Ans.2 A controlled electrical charge on a container of water will convert the water to hydrogen gas (H_2) and oxygen gas (O_2).

1515. In the treatment of cyanide poisoning which of the following is administered first?

1) Sodium thiosulfate
2) Sodium nitrite
3) Sulfur
4) Hyperbaric oxygen

Ans.2 Administration of sodium nitrite results in the formation of methemoglobin, which acts as a competitor of cytochrome a3 for cyanide. Since the affinity of methemoglobin for cyanide is greater than for cytochrome a3, the result is dissociation of the cyanide-cytochrome complex and resumption of oxidative metabolism. Sodium thiosulfate converts cyanide to thiocyanate.

1516. At a distance of 5 miles, an office building has an angle of elevation of 2º (neglecting the curvature of the earth). Find the height of the building.

1) 922 ft
2) 561 ft
3) 434 ft
4) 4,340 ft

Ans.1 5 miles is equal to 26,400 ft.
Draw a right triangle.
Tan 2º= Opposite side / Adjacent side
= Opp / 26,400 ft
Opposite side = tan 2º * 26,400 ft
= 922 ft

1517. An instantaneous exposure excursion to ammonia ranging up to 250 ppm should:

1) Be of no concern
2) Not be considered when the TWA is calculated
3) Be considered an unacceptable event
4) Be regarded as a measurement artifact

Ans.3 An excursion of this magnitude for a fast acting irritant would be considered unacceptable. For more information see the ACGIH TLVs and BEIs publication.

1518. Given a challenge atmosphere of 150 ppm and a quantitative fit test result of 3 ppm measured inside the respirator, what is the protection factor (PF) for that respirator?

1) 0.02
2) 3
3) 50
4) 147

Ans.3 To calculate the PF:
PF = outside concentration / inside concentration
PF = 150 ppm / 3 ppm
= 50

1519. A closed 500 l cylinder holds a gas at 200 psi(a) at 20ºC. If the cylinder is involved in a fire and increases in temperature to 260ºC, what is the gas pressure in the cylinder assuming an ideal gas?

1) 213 psi
2) 364 psi
3) 465 psi
4) 2,600 psi

Ans.2 $(P_1 V_1) / T_1 = (P_2 V_2) / T_2$
Remember to use absolute temperatures.
(200 psi * 500 l) / 293ºK = (P_2 * 500 l) / 533ºK
P_2 = 364 psi

1520. A chemical with a molecular weight of 88 evaporates from a chemical process. Where would it most likely accumulate?

1) Floor
2) Ceiling
3) Neither; it will mix uniformly
4) Insufficient information

Ans.1 For gases, density is proportional to molecular weight. Air has an approximate molecular weight of 29. Therefore the gas should sink.

1521. The reaction of oxygen and organic compounds yielding sudden and rapid combustion is known as a:

1) Detonation
2) Deflagration
3) Hypergolic reaction
4) Misantropic reaction

Ans.3 A hypergolic reaction is a violent reaction occurring when a fuel and oxidizer are mixed. The mixing of liquid hydrogen and liquid oxygen is also an example of this reaction.

1522. If the velocity of a reaction through the reacting material reaches sonic or supersonic speed, the reaction is called a:

1) Low order explosion
2) Low explosive
3) Deflagration
4) Detonation

Ans.4 The explosion that results when the velocity of reaction through the reacting material reaches sonic or supersonic speed is called a detonation.

1523. A sudden and violent release of large amounts of gas is called a(n):

1) Detonation
2) Hazardous reaction
3) Deflagration

4) Explosion

Ans.4 An explosion is the sudden and violent release of large amounts of gas.

1524. Which hydrocarbon has the lowest auto-ignition temperature?

1) Methane
2) Butane
3) Ethane
4) Decane

Ans.4 Decane is the hydrocarbon with the lowest auto-ignition temperature.

1525. Flammable limits and range is defined as:

1) The range in which flammable liquids cannot ignite
2) The range in which flammable liquids can auto-ignite
3) The range in which liquids form flammable mixtures with maximum concentrations of vapor in air or oxygen
4) The range in which gases or vapors form flammable mixtures with air or oxygen

Ans.4 Flammable limits and range is defined as the range in which gases or vapors form flammable mixtures with air or oxygen. There is a minimum concentration of vapor in air or oxygen below which propagation of flame does not occur on contact with a source of ignition.

1526. The vapor-air density is defined as:

1) The weight of a vapor-air mixture resulting from the vaporization of a flammable liquid at equilibrium temperature and pressure conditions as compared with the weight of an equal volume of air under the same conditions
2) The weight of a volume of a pure gas compared to the weight of an equal volume of dry air at the same temperature and pressure

3) The density of a solid or liquid substance is the ratio of the weight of the substance to the weight of the same volume of another substance
4) A hazard which exists wherever material that will burn or oxidize readily is available

Ans.1 The vapor-air density is defined as the weight of a vapor-air mixture resulting from the vaporization of a flammable liquid at equilibrium temperature and pressure conditions as compared with the weight of an equal volume of air under the same conditions.

1527. Methane (CH_4) is composed of carbon (atomic weight of 12.011 g/mol) and hydrogen (atomic weight of 1.008 g/mol).What is the molecular weight of methane?

1) 4.032 g/mol
2) 12.011 g/mol
3) 13.019 g/mol
4) 16.043 g/mol

Ans.4 One carbon atom (12.011 g/mol * 1) + four hydrogen atoms (1.008 g/mol * 4) = a molecular weight of 16.043 g/mol.

1528. Methane (CH_4) is composed of carbon (atomic weight of 12.011 g/mol) and hydrogen (atomic weight of 1.008 g/mol).What is the specific gravity of methane?

1) 0.41
2) 0.45
3) 0.55
4) 16.04

Ans.3 Molecular weight of methane is:
MW = (1 * 12.011 g/mol) + (4 * 1.008 g/mol)
= 16.043 g/mol
SG = MW of substance / MW of air
= (16.043 g/mol) / (29 g/mol)
= 0.5532

(A molecular weight of 29 is used for air. This is the weighted average of the gaseous components of air.)

1529. Methane (CH_4) is composed of carbon (atomic weight of 12.011 g/mol) and hydrogen (atomic weight of 1.008 g/mol).What is a primary concern, from a fire safety viewpoint, in large spills of liquid methane?

1) The gas has a specific gravity of less than 1.0 which will cause it to rise rapidly
2) The gas will spread over a wide area with little dilution
3) The gas has no odor
4) The gas will dilute quickly to reach its explosive concentration

Ans.2 Even though the specific gravity is less than one, when spilled in large quantities, the gas is cryogenic (cold) making it heavier than air. As the gas warms, it will spread over a wide area with little dilution.

1530. Find the vapor-specific gravity at 100ºF and atmospheric pressure for a flammable liquid which has a specific gravity of 2.3, and has a vapor pressure at 100ºF of 3.0 in Hg.

1) 1.1
2) 1.3
3) 1.5
4) 3.0

Ans.1 vasg = (ps)/P + (P - p)/P
Where:
p = vapor pressure of the liquid
s = specific gravity of the liquid
P = the pressure of one atmosphere in the same units as "p"
1 atm = 29.92 in Hg
vasg = [(3.0 in Hg * 2.3) / 29.92 in Hg] + [(29.92 in Hg - 3.0 in Hg) / 29.92 in Hg] = 0.2306 + 0.8997 = 1.1303

1531. How many kilowatts are required to raise 1 lb of water 1ºF in 1 sec?

1) 1,054 kW
2) 105.4 kW
3) 10.54 kW
4) 1.054 kW

Ans.4 1 Btu is the heat required to raise 1 lb of water 1ºF.
1 Btu = 2.930×10^{-4} kW-hr
There are 3,600 seconds in an hour.
$(2.930 \times 10^{-4}$ kW-hr)(3,600 sec/hr) = 1.054 kW

1532. An explosion of hydrogen in a battery recharging area is an example of a propagating reaction. Such a reaction is always:

1) Slow to grow
2) Endothermic
3) Exothermic
4) Spontaneous

Ans.3 Propagating reactions always require a source of ignition and are therefore exothermic.

1533. The term combustion usually refers to an exothermic, or heat producing, chemical reaction of a substance and:

1) Hydrogen
2) Oxygen
3) A substance with a similar molecular weight
4) An inert substance

Ans.2 Oxygen is required to support conventional combustion

1534. Water reactive materials are:

1) Materials that are chemically reactive with water
2) Materials that are chemically reactive with water while being heated
3) Materials that are chemically reactive with water under pressure
4) Materials that are chemically reactive with water while being heated under pressure

Ans.1 Water reactivity describes the sensitivity of materials to water without requiring heat or confinement.

1535. Decontamination procedures include all of the following EXCEPT:

1) Incineration
2) Chemical removal
3) Physical removal
4) Combination removal

Ans.1 Incineration is a form of disposal, not decontamination.

1536. The decontamination stations should be set up in:

1) The hot zone
2) The warm zone
3) The cold zone
4) Between the warm zone and the hot zone

Ans.2 Decontamination stations should be set up in the warm zone, which is the area between the contaminated area (hot zone) and the cold zone, where contamination should not occur. The decontamination line is the last point where decontamination will occur, at the entrance to the cold zone from the warm zone.

1537. In responding to a spill of hazardous materials, which of the following is NOT a good source of information for prompt hazard recognition?

1) MSDS
2) Hazardous waste manifest
3) Manufacturer label
4) Symbolic labels

Ans.2 The hazardous waste manifest does not contain hazard information.

1538. If the identity of the material in a pipe were unknown, one could determine the hazard of the pipe material by referring to all of the following EXCEPT:

1) MSDS
2) Process diagrams
3) Standard operating procedures
4) Other employees familiar with the processes

Ans.1 The MSDS can only be used if the identity is already known.

1539. The skill of prompt hazard recognition is so important because:

1) It allows quick spill clean-up
2) The responders can protect themselves and others
3) It allows plant management to understand the situation
4) Communication with the media is most efficient

Ans.2 The first concern is the health and safety of the responder.

1540. Under what circumstances would you want to designate a decontamination team for a spill response?

1) Only with spills greater than 55 gallons
2) Only with spills greater than 25 gallons
3) In practically every spill situation
4) Only in catastrophic spills greater than 1,000 gallons

Ans.3 In very simple defensive measures, decontamination may not be necessary, but typically at least one person should be assigned this responsibility for spill response situations.

1541. The first physiological signs of exposure to an oxygen-deficient atmosphere are increased rate and depth of breathing. This condition is called:

1) Anoxia
2) Hypoxia
3) Oxygen starvation
4) Decompression illness

Ans.2 Hypoxia, the first physiologic sign of an oxygen deficiency, is characterized by an increased rate and depth of breathing.

1542. The first consideration when you are the first person approaching an active spill is:

1) Stop the spill
2) Contain the spill
3) Move the drum
4) Personnel safety

Ans.4 The first consideration for first responders is their own safety and that of others. They are to determine what needs to be done to protect people.

1543. Supplied air respirators (SAR's) can be used in IDLH atmospheres under any of the listed conditions EXCEPT:

1) Employee is equipped with auxiliary escape air
2) Supply line are less than 300 ft in length
3) Maximum inlet air pressure is less than 125 psi(g)
4) Airflow into the face piece is 3 cfm

Ans.4 NIOSH respirator requirements require a face piece air flow rate of 4 cfm.

1544. A respirator that uses various chemical substances to purify inhaled air which contains certain contaminant gases and vapors is defined as a:

1) Chemically reactive air-purifying respirator
2) CHP respirator
3) Chemical cartridge air-purifying respirator
4) Chemical air-purifying respirator

Ans.3 A chemical cartridge air-purifying respirator is a respirator that uses various chemical substances to purify inhaled air which contains certain contaminant gases and vapors.

1545. Protective garment materials have all of the following general considerations EXCEPT:

1) They are permeable to some degree
2) They are resistant to all chemicals
3) Some chemicals have no acceptable garment
4) No single material is resistant to all chemicals

Ans.2 There is no material that can protect against all chemicals due to the many different chemical forms that are used in today's manufacturing. All materials have some level of permeation.

1546. What is permeation?

1) A chemical's ability to dissipate
2) A chemical's ability to pass through a material
3) A chemical's ability to dissolve another material
4) A chemical's ability to pass through living tissue

Ans.2 Permeation is a chemical's ability to pass through a material.

1547. Chemical protective clothing (CPC) can introduce all of the following hazards to an emergency responder EXCEPT:

1) Heat stress
2) Visual limitations
3) Physical limitations
4) Carbon monoxide

Ans.4 CPC can cause discomfort to the wearer, especially heat stress, and many physiological stresses during use. An SCBA is worn which supplies Grade D breathing air.

1548. An oil refinery worker was attempting to repair a pipe from which gas had been escaping. He became dizzy, noticed a pressure in his head, and then passed out. A fellow worker who went to his aid was also rendered unconscious. Both victims began convulsing and were hospitalized. The first victim appeared to be recovering but at approximately 72 hours after the exposure, developed severe pulmonary edema and died. The second victim recovered completely within 36 hours and was back at work within a week without lasting effect. What is the most logical chemical agent to cause this?

1) Hydrogen sulfide
2) Volatile organics
3) Carbon monoxide
4) Arsine

Ans.1 The most logical choice, given the fact that the incident takes place in an oil refinery, is hydrogen sulfide poisoning. Hydrogen sulfide interferes with cytochrome oxidase leading to tissue hypoxia.

1549. A semiconductor operator is exposed to a variety of agents when a diffusion furnace tube explodes during a process step. Upon admittance to the emergency room, the unconscious worker is found to have a bronze hue to his skin. Subsequent lab test indicate a number of lysed red blood cells. What is the causative agent?

1) Volatile organics
2) Arsine
3) Phosphine
4) Hydrogen chloride

Ans.2 The lysed red blood cells and bronze hue of the skin is highly indicative of arsine poisoning. This gas breaks apart the red blood cell. It is an extremely toxic chemical.

1550. Several young male agriculture pesticide formulation workers complained to the company physician about an inability to have children. Medical examination revealed little or no sperm and the decrease or loss of spermatogenic cells. What is the likely chemical cause?

1) Warfarin
2) DDT
3) DBCP
4) Aldrin

Ans.3 1,2-dibromo-3-chloropropane (DBCP) is a potent male reproductive toxin. It is also a suspect human carcinogen.

1551. A biotechnology technician dropped a bottle of compound X and saturated his right leg, the right side of his stomach and his chest. Fellow employees removed his shirt and washed his upper body. The technician then walked across the street to a

physician's office. However within 15 minutes of arriving, the technician collapsed and died. Subsequent autopsy revealed first and second degree burns on his body's right side. There was edema of both lungs, kidneys, pancreas, and spleen. What was the agent?

1) Hydrochloric acid
2) Phenol
3) Trichlorethylene
4) Deionized water

Ans.2 Phenol is a systemic poison absorbed through the skin. The removal of the shirt and subsequent washing of the chest did not remove the phenol from the victim's lower body and a subsequent acute poisoning occurred.

1552. How long should eyes or skin be flushed immediately after exposure to a hazardous chemical?

1) 3 min
2) 5 min
3) 10 min
4) 15 min

Ans.4 A general rule is to wash with copious amounts of water for at least 15 minutes.

1553. Which group of specially trained employees, who have been delegated by their employer, are most qualified to respond to an emergency in which a drum of hazardous substance has ruptured?

1) Paramedics
2) Nurses
3) HAZMAT team
4) First aid team

Ans.3 The HAZMAT team has been given the proper training that covers the necessary information to control and handle chemical substances, spills, and leaks safely, including information on the proper personal protective equipment and procedures that safeguard

employees against hazards and toxic substances exposure. In this example, the only choice that explicitly has the required knowledge would be the HAZMAT team - hazardous materials response team.

1554. What type of canister respirator would you use if you were working in a parking garage with a gas fired engine?

1) Blue
2) Black
3) Green
4) Red

Ans.1 Gas fired engines emit carbon monoxide as a by-product; the correct canister color for carbon monoxide is blue. Black is for organic vapors, green is for ammonia, and red is for all types of contaminants.

1555. The movement of liquids or vapors through a garment material that occurs on a molecular level is called:

1) Penetration
2) Osmosis
3) Convection
4) Permeation

Ans.4 Permeation involves sorption of the chemical into the surface of the material, diffusion through the material, and desorption on the opposite side. Penetration is the passage of vapors or liquids through zippers, seams, holes, etc.

1556. A one pound cylinder of Cl_2 (chlorine) fell off a lab bench and broke. The contents leaked into a sealed room of 60 ft x 30 ft x 15 ft. What is the concentration if the lab is at STP?

1) 20.5 ppm
2) 187.5 ppm
3) 2,050 ppm
4) 20,500 ppm

Ans.2 First calculate the volume of the room.
Vol = 27,000 ft^3 (one ft^3 is equal to 28.3 l)
= 764,640 l
Next calculate the mass of chlorine released.
1 lb Cl_2 = 454.5 g Cl_2
The molecular weight of Cl_2 is 71 g/mol
454.5 g(1 mol/71 g) = 6.4 mol
6.4 moles(22.4 l/mole) = 143.4 l Cl_2
ppm = Va/Vb x 10^6 (gives the concentration in ppm where Vb is the volume of the room)
= (143.4 l Cl_2 / 764,460 l) x 10^6 ppm/l
= 187.5 ppm Cl_2

1557. What is the dose of a chemical called that will kill 100% of the test animals?

1) LD_{50}
2) LD_{100}
3) LED_{100}
4) LET_{100}

Ans.2 The common terminology for this chemical dose is LD_{100} (i.e., the lethal dose for 100% of animals tested).

1558. The minimum temperature at which a liquid gives off sufficient vapor to form an ignitable mixture with air near the surface of the liquid is called the:

1) Auto ignition temperature
2) Flash point
3) Ignition point
4) Fire point

Ans.2 The auto ignition temperature is the temperature at which the liquid will self ignite with no external spark, and is used to determine the flash point. The ignition point is not a

"defined" fire term. The fire point is similar to the flash point except that the flammable substance is able to sustain self burning without a continued spark or ignition source.

1559. Reducing the oxygen concentration and raising the heat capacity of the atmosphere is the primary extinguishing mechanism of:

1) Inert agents
2) Halocarbon agents
3) Halon agents
4) Wet chemical agents

Ans.1 Carbon dioxide also works like inert agents to extinguish a fire. Halocarbons work through inhibiting the chain reaction of the fire process. Halon is banned for new installations. Wet chemicals inhibit the chain reaction of a fire and produce an emulsion effect that smothers the fire.

1560. Inside a flammable liquid storage room, the minimum portable fire extinguisher requirement is:

1) A 20 lb carbon dioxide adjacent to any dispensing station within the room
2) A multipurpose ABC dry chemical, located inside the room, adjacent to the entrance door
3) A fire hose, with fog nozzle, mounted on a column, inside the room
4) extinguisher must be located outside the room

Ans.4 In a hazardous location of this nature, the recommended practice is for the fire extinguishing equipment to be located outside the room.

1561. The maximum travel distance to an exit in an industrial occupancy fully protected with automatic sprinklers is:

1) 50 ft
2) 150 ft
3) 200 ft
4) 250 ft

Ans.4 The travel distance to at least one exit shall not exceed 200 ft in industrial occupancies NOT equipped with a sprinkler system or exceed 250 ft in industrial occupancies protected throughout by an approved, supervised sprinkler system. NFPA 101 40.2.6.1 (Life Safety Code)

1562. For a dust explosion to occur in a storage silo, which of the following conditions must be met:

1) Oxygen must be present in a concentration that will sustain rapid combustion
2) The dust must be well mixed with the oxygen at a concentration above the LEL
3) An ignition source must be present
4) All of above conditions must be met

Ans.4 A dust explosion compares with a deflagration in which gases spread through the burning material by conduction, convection, and radiation at a rate less than the velocity of sound. For it to occur, all conditions, including enclosed space, must be met.

1563. Class "B" fire doors are used to protect openings in:

1) Corridor walls
2) Stairwells
3) Fire walls
4) Exterior walls

Ans.2 Class B openings are in enclosures of vertical communication through buildings, such as stairs and elevators. Doors for protection of these openings have a fire protection rating of 1 to 1.5 hours.

1564. "Bonding" a flammable liquid dispensing can refers to the process of:

1) Connecting it to the earth
2) Attaching it to a grounding strap
3) Gluing it to the floor
4) Connecting it to other conductive objects

Ans.4 "Bonding" refers to the connection of conductive objects by means of a conductor to minimize potential electrical differences between the objects. "Grounding" is the process of connecting the conductive objective to the ground. In flammable liquid dispensing, both "Bonding" and "Grounding" are important.

1565. A detonation has the following features EXCEPT:

1) Pressure front moves supersonically
2) Potential for extreme destruction exists
3) Blowout panels and rupture disks can vent explosion
4) Blowout panels and rupture disks are useless

Ans.3 Due to the supersonic nature of the pressure front after a detonation, blowout panels and rupture disks are useless.

When materials explode they usually propagate as a deflagration of a detonation. A deflagration is a low speed propagation and is effectively an expansion of gases. An expansion or flow of gases can be directed through venting and thus serious damage can be prevented through adequate venting. A detonation, on the other hand, results in a shock wave that cannot be redirected. The shock wave will impart its energy on any surface that is in its path, thus venting is practically useless for protection against damage from a detonation.

National Fire Protection Association, Fire Protective Handbook, (15th ed.) Section 3/Chapter 2 subparagraph A.

1566. You are a safety professional for a company that occupies several three-floor office buildings at its corporate headquarters location. Life safety is a concern, and the company insists that recognized life safety standards are not only enforced at its high-hazard plants, but also deployed across all corporate headquarters facilities. What is the minimum width permitted for the common corridors and passageways that lead to exits in the corporate headquarters buildings?

1) 28 in
2) 30 in
3) 36 in
4) 44 in

Ans.4 44 in. is the minimum permitted width of corridors and passageways that lead to means of egress. NFPA 101 39.2.3.2 (Life Safety Code)

1567. Your office area in your building is about 10,000 ft^2. During an office inspection, someone asks you how many fire extinguishers are needed. The current extinguishers in place are class 2A extinguishers. What is your answer?

1) 1
2) 2
3) 4
4) 6

Ans.3 Four are needed, since each one can cover 3,000 ft^2 in an office occupancy.

1568. There are different types of standpipe systems in buildings. Type I systems are meant to be used by:

1) Sprinkler systems
2) Fire department personnel
3) Building occupants
4) Fire brigades

Ans.2 These systems are for use by the fire department personnel.

1569. One of the specific concerns for fighting fires where oxidizing agents are present is that these agents:

1) Inhibit chemical reactions
2) Provide oxygen for the fire
3) Limit the use of water as an extinguishing agent
4) Are highly combustible

Ans.2 These agents provide the fire with a source of oxygen even when extinguishing agents are used.

1570. Assume you are building a five-story building and will place the heating unit in the basement. If the vent is to pass through the floors and roof, what must you do to the vent?

1) Install fire suppression equipment in the vent
2) Increase the size of the vent by a factor of 1.2
3) Install the vent in one-hour rated fire resistive walls
4) Install the vent in two-hour rated fire resistive walls

Ans.4 Two hour walls are needed when the building is four stories or more in height.

1571. When water supply pipes change direction due to installation requirements, the installer must often use what technique to prevent damage to the system when in use?

1) Use of plastic or rubber pipe
2) The installation of thrust block
3) Shear plates
4) Shock absorber pistons

Ans.2 Thrust blocks are used at tees, plugs, and bends to absorb loads imposed by water moving in the pipe.

1572. In normal areas the spacing between fire hydrants should not exceed:

1) 500 ft
2) 800 ft
3) 1,000 ft
4) 1,600 ft

Ans.2 In normal areas, 800 ft is the normal spacing. In congested areas, 500 ft may be used. NFPA Fire Protection Handbook

1573. How often should private hydrants be opened and closed to ensure proper water flow and drainage?

1) Monthly
2) Semi-annually (twice per year)
3) Annually
4) Bi-annually (once every two years)

Ans.3 Annual flow testing is recommended for private hydrants. NFPA 25 - Standard for the Inspection, Testing, and Maintenance of Water-Based Fire Protection Systems

1574. Your company is planning a new corporate office and training center. This center will include a large meeting room that can be subdivided into smaller meeting rooms with movable walls. What is one of the commonly overlooked items when designing such facilities?

1) The facility needs to be classified as a place of assembly
2) Emergency exit provisions must be made for each area subdivided by movable walls
3) The facility may be used by the public
4) Generally these facilities have cooking facilities that must be treated as a restaurant

Ans.2 Often overlooked is the need to provide emergency egress from areas subdivided by movable walls, particularly when several scenarios of subdivision are possible.

1575. All of the following statements are true EXCEPT:

1) Dust-ignition proof equipment is designed to keep ignitable amounts of dust from entering the enclosure

2) Dust may accumulate on electrical equipment, causing overheating of the equipment, as well as the dehydration and gradual carbonization of organic dust deposits
3) Overheated equipment may malfunction and cause a fire
4) Equipment must be designed to operate at the ignition temperature of the specific dust involved even when blanketed

Ans.4 Equipment must be designed to operate below the ignition temperature of the specific dust involved even when blanketed.

1576. Ignition of flammable vapors, dusts, or fibers by discharge of static electricity can be prevented by any of the following EXCEPT:

1) Grounding and bonding
2) Insulation
3) Humidification
4) Ionization

Ans.2 Static electricity is non-flowing electricity commonly generated by friction and the effects of the positive and negative charges which have the potential to spark. The static charge which already exists can be removed or allowed to dissipate itself. The acts or conditions which can accomplish this are the same as those which permit the separation of charges to occur. Ignition is prevented by the actions employing humidification, ionization, grounding, and bonding. Flammable vapors, dusts, or fibers cannot be insulated.

1577. In Class II hazardous locations there are three groups: E, F, and G. Special designs are required to prevent dust from entering into the electrical equipment enclosure. All of the following are true EXCEPT:

1) Assembly joints and motor shaft openings must be tight enough to prevent dust from entering the enclosure
2) The design must take into account the insulating effects of dust layers on equipment and must ensure that the equipment will operate below the ignition temperature of the dust involved
3) If conductive combustible dusts are present, the design of equipment must take the special nature of these dusts into account
4) Equipment that is dust-ignition proof is generally not acceptable for use in Class III locations

Ans.4 Equipment that is dust-ignition proof is generally acceptable for use in Class III locations, since the same design considerations are involved. NFPA 70 Article 500-3 (National Electric Code)

1578. As a safety professional, you are asked to recommend a fire detector for use in a Class I, Division 2 work area. Which of the following would you select?

1) Fixed temperature
2) Rate compensated thermal
3) Rate-of-rise thermal
4) Bulb detection system

Ans.4 The bulb detection system is completely mechanical thereby avoiding the use of electricity and a potential source of ignition in an area where volatile flammable liquids or gases are used.

1579. An average of one gallon of toluene is evaporated every hour in a continuous process drying oven operating at sea level. Assuming that the toluene completely mixes throughout the oven, what is the minimum exhaust ventilation rate required by code to prevent an explosive concentration? The code does not permit the concentration of flammable vapors in the oven to exceed 25% of the LEL at any time.

The LEL for toluene is 1.1% at the oven's operating temperature
The specific gravity of toluene is 0.87 (water is 1.0)
The specific vapor density of toluene is 3.1 (dry air is 1.0)
The density of water at STP is 8.328 lb/gal
The density of air at STP is 0.075lb/ft^3

The basic equation to determine the required ventilation to maintain a system at a discrete LEL is:

Ventilation Rate = (Density of Water at STP)/(Density of Air at STP)(Specific Gravity of Solvent)/(Specific Vapor Density of Solvent) [(100% - LEL)/LEL] Solvent Evaporation Rate

1) 31 SCFM
2) 47 SCFM

3) 187 SCFM
4) 2,802 SCFM

Ans.3 First, realize that the given equation gives you the required ventilation to MAINTAIN the LEL. We need to know the required ventilation rate at 25% of the LEL. Therefore, we will have to multiply the result from the given equation by a factor of 4 (100% of LEL/25% of LEL = 4). Also note that we require the answer in standard cubic feet per MINUTE. Since the evaporation rate is gallons per hour, we will have to convert to gallons per minute. Carefully track the units when solving this equation, and remember that specific gravity and specific vapor density have no units.

Minimum Rate = 4(8.328 lb/gal / 0.075 lb/ft^3)(0.87 / 3.1) [(100% - 1.1%) / 1.1%] (1 gal/hr) (1 hr/60 min)
Minimum Rate = 4 * 111.04 ft^3/gal * 0.281 * 89.91 * 0.0167 gal/min
Minimum Rate = 187 ft^3/min

NFPA 86 Section 7 - Standards for Ovens and Furnaces

1580. The water level of a rectangular fire pond, 75 ft * 42 ft, is to be raised 4 in.. How many gallons are needed?

1) 7,875 gal
2) 15,750 gal
3) 23,625 gal
4) 31,500 gal

Ans.1 V = 75 ft * 42 ft [4 in / (12 in / 1 ft)]
= 1,050 ft^3
There are 7.5 gallons in 1 cubic foot (ft^3)
V = 1,050 ft^3 * 7.5 gal/ft^3
= 7,875 gal

1581. Water spray is an acceptable extinguishing method for all of the following EXCEPT:

1) Alcohol liquid fires
2) Sodium fires
3) Oil-filled transformer fires
4) Corrugated box fire

Ans.2 The extinguishing properties of water include cooling, smothering, emulsifying, and diluting. The application of water in spray form or combination nozzles and sprinklers is considered to be an efficient method for extinguishing fire. However, using water to extinguish a burning metal fire may cause an explosion.

1582. Halon 1301 is a clean extinguishing agent that leaves no residue. It has a "B" and "C" extinguishing classification, intended primarily for use on sensitive electronic equipment such as computer operations. In a 10,000 ft^2 computer room, what concentration of Halon 1301 would be needed to prevent combustion?

1) 2%
2) 6%
3) 8%
4) 15%

Ans.2 For Halon 1301 to be effective, the ventilation systems must be stopped so that the designed concentration of 6% will effectively extinguish the fire. In a total flooding system, human exposure should be limited to one minute if the concentration is as high as 7% to 10%.

1583. It is considered good practice to avoid all unnecessary exposure to Halon 1301 and to prohibit exposure above:

1) 15%
2) 18%
3) 20%
4) 25%

Ans.1 It is considered good practice to avoid all unnecessary exposure to Halon 1301 and to prohibit exposure above 15%.

1584. On a loading dock in Minnesota, you would most likely find the following type of sprinkler system:

1) A wet pipe system
2) A halon system
3) A dry-pipe system
4) No sprinklers

Ans.3 Dry-pipe systems are used where there is a possibility of freezing.

1585. In a fire sprinkler system, what is the purpose of a check valve?

1) To limit water pressure to the sprinkler system
2) To check the flow and send an alarm to the central station
3) To prevent water from entering a dry pipe system
4) To prevent the sprinkler system from flowing back into a public water system

Ans.4 In some cases due to health requirements, two check valves may be needed in a system, but normally one is sufficient to prevent sprinkler system water from returning to the public water supply system.

1586. If an SCBA is worn in an atmosphere that is immediately dangerous to life and health; he/she must:

1) Wear safety harnesses and lifelines
2) Have a standby worker with a canister respirator
3) Have an escape pack
4) Not go into the area

Ans.1 In the event of an emergency, a safety harness and lifeline is the means of quickly removing the person from the hazardous atmosphere.

1587. What type of breathing air is used for supplied air respirators?

1) Class A
2) Class B
3) Class C
4) Class D

Ans.4 Class D breathing air is considered safe for supplied air respirators. Class D breathing air has specific levels of oxygen, hydrocarbons, carbon monoxide, and carbon dioxide.

1588. Which of the following SCBA systems is required for IDLH atmospheres?

1) Demand
2) Negative pressure demand
3) Pressure demand
4) Gas mas

Ans.3 Pressure demand SCBA's provide a continuous flow of positive pressure air to the face piece of the respirator. The demand/negative pressure demand respirators supply air to the face piece only upon inhalation. Gas masks are not SCBA's.

1589. Class B portable fire extinguishers are rated by the size of the fire that a novice can extinguish and are identified by a red square containing the letter "B". The numeral preceding the "B" in the rating indicates:

1) Number of seconds it will take to dispense the contents of the extinguisher
2) Number of times the extinguisher may be used
3) Number of square feet of burning liquid that can be extinguished
4) Number of times the unit is effective compared with one gallon of water

Ans.3 The numeral preceding the letter "B" indicates the number of square feet of burning liquid that the novice can extinguish with that extinguisher. For example a 20B rating indicates the novice can extinguish 20 ft^2 of burning liquid fire.

1590. Class A portable fire extinguishers are identified by a green triangle containing the letter "A." What does the rating number appearing before the "A" signify?

1) The time in seconds it takes to dispense the contents of the extinguisher
2) The time in minutes it takes to dispense the contents of the extinguisher
3) The ability to extinguish a fire as compared to one gallon of water
4) e number of square feet of paper waste that can be extinguished

Ans.3 The number preceding the letter "A" rates its ability to extinguish a fire as compared to one gallon of water. For example, a "4A" rating means the unit is four times as effective as one gallon of water, or conversely, it has the same effectiveness as four gallons of water.

1591. As a safety professional, you must recommend how to safely store the following quantities of flammable and combustible liquids: Class I - 35 gallons Class II - 35 gallons Class III - 50 gallons

1) Provide two (2) approved metal storage cabinets
2) Disburse the materials throughout the assembly area
3) Provide one (1) approved metal storage cabinet
4) Insist the quantities be reduced

Ans.1 NFPA 30 states that not more than 120 gallons of Class I, Class II and Class IIIA liquids may be stored in a storage cabinet. Of this total, not more than 60 gallons may be of Class I and Class II liquids.

1592. In a 25 ft high assembly area designed for fire safety, a 6 ft metal curtain has been affixed to the bar joists. What is the LEAST likely fire safety purpose?

1) Delays and limits the horizontal spread of heat, gases, and smoke
2) Assists in making the venting more effective
3) Provides a horizontal confinement to obtain a stack effect
4) Delays and limits the vertical spread of heat, gases, and smoke

Ans.4 The metal curtain is a draft curtain that functions to delay and limit the horizontal spread of heat, gases, and smoke by providing a horizontal confinement to obtain a stack

effect. The placement of draft curtains makes the venting more effective in that the volume of hot gases and smoke discharged by thermal updrafts through a vent varies directly with the height of the gas or smoke column.

1593. The term central station alarm or central station system is used to describe a type of alarm system that communicates trouble at a premises to those who will respond. These systems:

1) Directly notify the local fire department
2) Notify a constantly attended location operated by a company that notifies appropriate authorities
3) Notifies emergency personnel that work at the facility
4) Notifies a central point in the facility so that a designated person can call the fire department

Ans.2 This type of alarm system notifies a continuously attended facility operated by a company whose purpose is to alert appropriate authorities.

1594. The best fire protection system advisable in a plant where water damage would result in a serious product loss would be:

1) Pre-action
2) Deluge
3) Dry pipe
4) Wet pipe

Ans.1 Pre-action fire protection systems are designed to eliminate the danger of water damage and to counteract the operational delay of conventional dry pipe systems. In pre-action systems, the water supply valve is actuated independently of the opening of sprinklers.

1595. Confronted with having to extinguish a small Class A fire in an office where only the following extinguishing agents were present, which would you select?

1) Bucket of sand
2) Multipurpose dry chemical
3) Carbon dioxide

4) Nothing (let it burn out)

Ans.2 A multipurpose dry chemical is effective because it provides a rapid knock-down of flames and forms a fire-retardant coating that prevents reflash. CO_2 may blow the contents around, leaving embers which can reflash.

1596. Which class of fire extinguisher is best for extinguishing a paper fire?

1) Class "A" (green triangle with "A" in triangle)
2) Class "B" (red square with "B" in square)
3) Class "C" (blue circle with "C" in the circle)
4) Class "D" (yellow star with "D" in the star)

Ans.1 Class "A" fire extinguishers are used on paper or wood fires. Class "B" fire extinguishers are used on burning flammable and combustible liquids. Class "C" fire extinguishers are used on fires involving energized electrical equipment. Class "D" fire extinguishers are used on fires involving combustible metals.

1597. Spark arresters are required on chimneys in certain applications. In which of the following applications is a spark arrester not required but is desirable?

1) On refuse burners
2) On incinerators
3) On solid fuel furnaces
4) On all heating equipment burning liquid fossil fuels

Ans.3 Spark arresters are required on refuse burners and incinerators. Arrestors are desirable (but not required) when burning solid fuels. They are not needed on liquid or gas fuels.

1598. The best way to extinguish a chimney fire serving a wood burning stove or heater is to:

1) Close the damper and doors reducing the air supply
2) Let them burn out
3) Use water in the fire box
4) Use water in the chimney

Ans.1 Close the damper to cut off oxygen to the fire. If it does not totally cut off the air, it will reduce the intensity of the fire.

1599. One of the most effective forms of area protection from fires in combustible oil quenching operations is:

1) Draining the oil to a tank
2) Foam systems
3) Automatic water spray systems over the tanks
4) Automatic closing lids

Ans.3 The best protection is water sprays since they reduce heat, reduce oxygen, and protect the building from oil fires. The other controls can also be used but are not generally as effective as water spray.

1600. The mixing room for a spray painting operation should be located:

1) In the flammable liquids storage room
2) In the spray painting room or booth
3) Adjacent to an outside vented wall
4) Outdoors

Ans.3 The best location is in a separate room located on an outside wall of the building so that explosion venting can be provided. It should be isolated from the rest of the building by fire-resistive construction.

1601. High piled combustible storage presents a challenge to sprinkler systems. Above what height are special in-rack sprinkler system design considerations necessary?

1) 6 ft
2) 8 ft
3) 12 ft
4) 16 ft

Ans.3 12 ft is considered the upper range for piled combustibles without utilizing special system design techniques that can get water to a potential fire.

1602. Your company is a grocery wholesaler and is planning a new warehouse with storage racks 60 ft high. What is one of the primary reasons you will recommend a sprinkler system that includes in-rack sprinklers?

1) In-rack sprinklers are required by code and will reduce the fire insurance costs
2) In-rack sprinklers are cheaper to install than other types of systems and are easy to protect from damage
3) Water requirements can be reduced when in-rack sprinklers are installed in addition to general building sprinklers
4) High rack storage of commodities on pallets and in boxes will create significant flues through which fire will propagate

Ans.4 In-rack storage creates flues in the storage area, particularly between pallets of goods stacked side-by-side. In-rack sprinklers will knock down the fire sooner and will probably reduce water damage from fire suppression. They will cost more to install but will protect the building and the contents.

1603. Who is LEAST responsible for ensuring that each container of a hazardous chemical shipped to a facility is labeled?

1) Chemical manufacturer
2) Importer
3) Distributor
4) End user

Ans.4 The chemical manufacturer, importer, or distributor must ensure that hazardous chemical containers are labeled.

1604. SCBA is:

1) Service connected binding arbitration
2) Society for certified breathing apparatus
3) Service certified breath analysis
4) Self-contained breathing apparatus

Ans.4 This is the common acronym for self-contained breathing apparatus.

1605. A truck driver transporting hazardous materials must know the following:

1) Rest stops along the intended route
2) First aid
3) Nature of cargo
4) CPR

Ans.3 The truck driver is legally obligated to know the nature of the cargo being transported.

1606. What type of respirator is best for initial response during an emergency?

1) Organic vapor/acid gas cartridge full-face air-purifying respirator
2) Powered air-purifying respirator
3) SAR with escape bottle
4) SCBA

Ans.4 A positive pressure SCBA is recommended until the type and level of hazardous chemicals have been identified.

1607. In responding to a spill, one would use SCBA in the following situations EXCEPT:

1) When the oxygen level is below 19.5%
2) When the concentration of the material is greater than the protection factor for other levels of respiratory protection
3) When the hazards or concentrations are unknown
4) When oxygen deficiency is not a factor, and the maximum concentrations of airborne hazardous chemicals are within the limits of appropriate NIOSH-approved air-purifying respirators

Ans.4 If oxygen deficiency is not a problem, then a properly selected air-purifying respirator may be used.

1608. Which group of specially trained employees, who have been delegated by their employer, is qualified to respond to an emergency in which a drum of hazardous substance has been ruptured?

1) ER Team
2) PEO Team
3) HAZMAT Team
4) Any of the above

Ans. All of the above groups, given the proper training that covers the necessary information to control and handle chemical substances, spills, and leaks safely, including information on the proper personal protective equipment and procedures that safeguard employees against hazards and toxic substances exposure, can respond to a ruptured drum of a hazardous substance.
ERT = emergency response team
PEO = plant emergency organization
HAZMAT = hazardous materials response team

1609. You are recommending that your organization purchase an expensive digital trunked radio system for emergency response to replace the single frequency system currently being used. Your manager is looking at several other systems with similar features and capabilities and wants to compare them from only a cost-basis. He has

asked you to give him the total cost of the system in today's dollars. Your company uses an annual interest rate (compounded annually) of 8.5% to perform these types of small capital project comparisons. Based on the data provided in the Table below, what is the total system cost assuming a useful life of 15 years? Note that the maintenance agreement and utility costs are assessed at the end of the first year.

Cost of 8 portable radios and one base station	$25,000
Cost of antenna and transmitter (and facilities)	$55,000
Cost of electrician labor and materials	$17,500
Cost of computer aided dispatch computers	$10,000
Annual maintenance agreement cost	$12,500
Annual utilities above those from current system	$2,500
Anticipated total system salvage value in 15 years	$7,500

1) $229,858
2) $234,270
3) $325,000
4) $414,375

Ans.1 This is a present value problem with an annuity and a future value.

The capital cost of the system is $25,000 + $55,000 + $17,500 + $10,000 = $107,500

The annual costs are ($12,500 + $2,500) for 15 years. The present value of this annual cost (annuity) is

$PV = A [(1 + i)^n - 1] / [i (1 + i)^n] = \$15,000 (1.085^{15} - 1) / (0.085 * 1.085^{15}) = \$124,564$

The present value of the salvage value of $7,500 is

$PV = FV (1 + i)^{-n} = \$7,500(1.085)^{-15} = \$2,206$

The TOTAL cost of this option is therefore the present value of all of the costs MINUS the present value of the salvage value.

Total Cost = $107,500 + $124,564 - $2,206 = $229,858

1610. What color is used to identify ionizing radiation?

1) Red
2) Magenta
3) Green
4) Orange

Ans.2 A magenta field identifies areas with ionizing radiation.

1611. Both DOT and NFPA systems use _____ coloring to identify an oxidizing material.

1) Red
2) Yellow
3) Green
4) Blue

Ans.2 Although systems employed by DOT and NFPA for identifying hazardous materials are similar, their purposes differ as does their color coding, with the exception of yellow for oxidizing materials and red for flammable materials.

1612. What does a "3" in the health section of a HMIS label mean?

1) Poison
2) Moderate health hazard - use reasonable precautions
3) Non-hazardous
4) Very hazardous to health - avoid all routes of exposure

Ans.4 The HMIS is the hazardous materials identification system commonly used to identify chemicals. The rating scale is 0-4, 0 being the least hazardous and 4 being the most hazardous. Materials are rated for reactivity, flammability, and health hazards

1613. The NFPA 704 diamond has four classes of materials by position and color on the diamond. The classes are:

1) Health hazard, flammability, reactivity, and specific hazards
2) Toxicity, ignitability, corrosivity, and reactivity
3) Explosivity, health, flammability, and specific hazards
4) Health hazards, flammability, reactivity, and corrosivity

Ans.1 The four classes of materials and colors on the National Fire Protection Association 704 (NFPA 704) diamond are:

- health hazard-blue
- flammability-red
- reactivity-yellow
- and specific hazards-white.

1614. HMIS labels contain the following information:

1) Health, fire, reactivity, toxicity
2) Health, fire, reactivity, personal protective equipment
3) Toxicity, fire, reactivity, personal protective equipment
4) Health, fire reactivity, special precautions

Ans.2 The Hazardous Materials Information System (HMIS) uses a standard label for communicating hazards on health, fire, reactivity, and recommended PPE.

1615. On vehicles used for transporting hazardous cargo, placards are used as a warning. A placarded vehicle must have at least how many identical placards?

1) One
2) Two
3) Four
4) Six

Ans.3 Placards are used to warn others of hazardous cargo. Placards are signs put on the outside of a vehicle to show the hazard class of the cargo. A placarded vehicle must have at least four identical placards. They are put on the front, rear, and both sides. Placards must be readable from all four directions. There are 19 DOT-specification placards. They are 10¾ inches square, turned upright on a point (in the shape of a diamond).

1616. A hazardous chemical shipment Bill of Lading must have the NFPA 704 label.

1) True

2) False
3) Depends on the type of hazard
4) Depends on the destination

Ans.2 The Bill of Lading must have the hazard class, chemical ID number, and the quantity of the transported chemical, but the NFPA 704 label is not required.

1617. Aspects of the NFPA 704 labeling system include all of the following EXCEPT:

1) May be used without a supplementary manual
2) Must list all chemical hazards
3) Warns of worst hazards possible
4) Easily understood

Ans.2 The NFPA 704 system is easy to understand, may be used without a supplementary manual, and summarizes the worst hazards.

1618. Disadvantages of the NFPA 704 labeling system include:

1) Gives minimal information
2) Not extensively used
3) Difficult to understand
4) Misleading

Ans.1 The summarized information on the NFPA 704 label is frequently inadequate for emergency response and other applications where detailed, accurate hazard information is essential.

1619. In the NFPA 704 diamond symbol, the top number indicates the degree of _____ risk of a certain chemical.

1) Health
2) Flammability
3) Reactivity
4) Water reactivity

Ans.2 The NFPA 704 diamond symbol is used to denote the health, flammability, and reactivity risk of a substance. The top number in the red-background area indicates the degree of flammability hazard, the left number in the blue-background area indicates the degree of health hazard, the right number in the yellow-background area indicates the degree of reactivity hazard, and the symbol in the lower, white-background area of the diamond indicates special hazards such as water reactivity, radioactivity, etc.

1620. Planning for spills, explosions, etc. is considered:

1) Job safety analysis
2) Containment planning
3) Emergency planning
4) Backing up

Ans.3 Emergency planning includes planning for spills, explosions, etc.

1621. Your company has an emergency brigade established under 29 CFR 1910.156. As your plant has expanded, the roles and capabilities of your company's fire brigade have also expanded. To promote community goodwill and to provide an opportunity to interact with the local volunteer fire department, your company is seeking an opportunity to provide its fire suppression, rescue, and emergency medical services to the community within the local fire district. What is a common method for outlining the roles, responsibilities, and communication channels between your company's fire brigade and the community fire district should this be agreeable to the community?

1) Fire fighting agreement
2) Mutual aid agreement
3) Agreement of contracted services
4) Emergency response agreement

Ans.2 These types of agreements are called mutual aid agreements. The terms of these agreements detail what (if any) compensation will occur as a result of a request for an emergency response by both parties as well as chains of command, communication channels, and other policy-related information.